

UNDERGRADUATE PROJECT PROPOSAL

Waste-Heat-Driven Water Recovery from Data Centre Cooling Systems

Thermal integration of a hybrid RO–membrane distillation island with a warm-water liquid-cooled data centre, with model-based dispatch and permeate quality verification.

DeccAI — working title of the integrated system

ABSTRACT

Indian data centres reject heat to atmosphere through evaporative cooling towers while drawing make-up water from aquifers that are, in the case of the Devanahalli block near Bengaluru, already being extracted at 169 % of the sustainable limit. The rejected heat is discarded at 40–55 °C and the concentrated tower blowdown is discharged to drain. This proposal treats both as recoverable streams.

The work develops a thermal design for a hybrid reverse-osmosis / membrane-distillation (RO–MD) recovery island driven by the facility's own cooling-loop return heat, integrated with an elevated-temperature (ASHRAE W4) direct-to-chip liquid cooling architecture. Membrane distillation is selected specifically for the high-salinity RO reject, where osmotic pressure makes further pressure-driven recovery uneconomic but vapour-pressure-driven transport is essentially unaffected.

A first-principles system model has been implemented and validated qualitatively against a real-time simulation console; the proposal now seeks to construct a bench-scale direct-contact MD rig to determine the membrane distillation coefficient B and the ion rejection achieved on synthetic Devanahalli-chemistry blowdown, and to calibrate the polarisation model. Expected outcome: a validated design method sizing an MD island against any facility's measured heat-rejection profile and blowdown chemistry.

DESIGN CASE

2.4 MW IT load · Bengaluru Rural

DURATION

8 months

DISCIPLINE FOCUS

Heat transfer · mass transfer · thermal systems

ESTIMATED BUDGET

₹9.45 lakh

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Headline design figures

2 304 kW Heat rejected at design IT load	52 °C Coolant return, ASHRAE W4	92.5 % Blowdown recovery, hybrid RO–MD	89 % Reduction in make-up water, full scheme
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1 Introduction and motivation

1.1 Thermal loads in a modern data centre

Essentially all electrical power delivered to information-technology (IT) equipment in a data centre is converted to heat. For a facility with an IT load of 2.4 MW, the heat that must be removed continuously is approximately 2.30 MW — comparable to the thermal duty of a small process plant, but rejected at a very low temperature and therefore of low thermodynamic value.

Two features distinguish this heat source from conventional industrial waste heat:

- **It is low-grade but extremely steady.** Unlike a batch furnace, a data centre rejects heat 8 760 hours a year, with diurnal modulation of roughly $\pm 30\%$ about the mean.
- **Its grade is a design choice, not a given.** The coolant supply temperature is set by the operator. Raising it from a conventional 18 °C chilled-water regime to an ASHRAE W4 warm-water regime at 40 °C simultaneously eliminates mechanical chilling and raises the return temperature into a range where the heat can actually do useful work. This is the single most important thermodynamic decision in the whole scheme.

1.2 Water consumption in evaporative cooling

Where heat is rejected through an open evaporative cooling tower, the rejection mechanism is predominantly latent. Water is consumed — permanently removed from the local catchment as vapour — and a second, smaller stream (blowdown) must be bled continuously to prevent the dissolved-solids concentration from rising to the point of scaling and corrosion.

The distinction between *withdrawal* and *consumption* matters and is frequently blurred in industry reporting. For the design case developed in §6.3, evaporation accounts for 75 % of make-up and is genuinely consumed; blowdown accounts for the remaining 25 % and is, in principle, entirely recoverable.

1.3 The Indian hydrological context

Table 1.1 — Context figures for the design location and national comparison.

QUANTITY	SIGNIFICANCE	SOURCE
169 %	Stage of groundwater extraction, Devanahalli block — the design location. Extraction exceeds annual recharge.	CGWB
$\approx 300\text{ m}$	Typical borewell depth in the block; natural recharge penetrates to roughly 60 m.	Field reporting
60.47 %	National stage of extraction; 11.1 % of assessment units classified over-exploited.	CGWB 2024
$> 1.5\text{ mg/L}$	Fluoride concentration in deep hard-rock aquifers of this geology, against the WHO drinking-water limit.	WHO; Springer 2026
20 → 50 %	Statutory wastewater reuse floor for large water users, 2027–28 rising to 2030–31.	LWM Rules 2024

The regulatory trajectory is therefore toward mandated reuse fractions that an evaporative facility cannot meet by improving tower efficiency alone. Recovery of the blowdown stream, and reduction of evaporation through architectural change, are both required.

2 Problem statement and research gap

2.1 Statement

PROBLEM

A liquid-cooled data centre simultaneously rejects 2.3 MW of low-grade heat to atmosphere and discharges approximately 1 m³/h of concentrated, contaminated blowdown to drain, while drawing make-up water from a stressed aquifer. No thermal integration exists between the two streams, although the rejected heat is at a temperature that can drive a thermally-activated separation and the blowdown is precisely the stream such a separation is suited to.

2.2 Why the blowdown stream rather than raw groundwater

Treating the intake would require a plant sized for the full make-up flow (3.9 m³/h) and would not reduce discharge. Treating the blowdown addresses a stream one quarter the size that is *already* concentrated — improving separation economics, because the driving force for any concentration-based recovery scales with feed concentration — and simultaneously eliminates a permitted discharge. It also requires no additional groundwater abstraction consent, which materially simplifies deployment.

2.3 Why a thermally-driven separation

Reverse osmosis is the default choice for brackish recovery and is retained in this design for the bulk duty. However, RO is pressure-driven: the applied pressure must exceed the osmotic pressure of the concentrate, which rises linearly with concentration. Waste heat is of no use to an RO unit whatsoever.

Membrane distillation (MD) is driven by the partial-pressure difference of water vapour across a hydrophobic microporous membrane. Its driving force depends on temperature and only weakly on concentration, through water activity. The two technologies therefore fail in opposite regimes, which is what makes the hybrid the correct configuration rather than a compromise:

Table 2.1 — Osmotic pressure and water activity along the concentration train (design case, §8).

STREAM	TDS MG/L	OSMOTIC PRESSURE BAR	WATER ACTIVITY A_w	PRACTICAL CONSEQUENCE
Tower blowdown	1 680	1.3	0.999	Well within RO capability
RO reject	6 720	5.2	0.996	RO recovery limited by LSI / scaling, not pressure
MD concentrate	22 400	17.5	0.992	RO would need > 30 bar feed; MD flux penalty is 0.8 %

The osmotic pressure rises by a factor of 13 across the train; the MD driving force falls by 0.8 %. This is the technical justification for the architecture and the central argument of the proposal.

2.4 Gap addressed

- Published MD work is dominated by seawater desalination at feed temperatures of 60–80 °C supplied by solar or dedicated boilers. Data-centre return heat is available at 45–55 °C, where flux is roughly a quarter of the seawater-desalination case. **Design methods for this grade of heat are not established.**
- Prior art for waste-heat-driven desalination in data centres (see §4.4) assumes a single steady liquid loop with a seawater feed. Indian facilities are inland, mixed air- and liquid-cooled, and thermally variable with IT load. **The transient behaviour is unaddressed.**
- Rejection performance of MD against **fluoride** at the concentrations characteristic of Karnataka hard-rock aquifers has limited published data at these operating temperatures.

3 Aim, objectives and scope

3.1 Aim

To develop and experimentally validate a thermal design method for a waste-heat-driven hybrid RO–MD water recovery island integrated with a warm-water liquid-cooled data centre.

3.2 Objectives

1. Establish the energy and mass balance for a 2.4 MW liquid-cooled facility under three cooling architectures, and quantify the make-up water demand of each.
2. Develop a resistance-in-series transport model for direct-contact MD incorporating temperature polarisation, and size the module against the available heat grade.
3. Design the heat-recovery exchanger linking the cooling loop to the MD feed circuit, including ϵ –NTU verification and pinch analysis.
4. Construct a bench-scale DCMD rig and determine the membrane distillation coefficient B and ion rejection for synthetic blowdown matching Devanahalli groundwater chemistry.
5. Calibrate the system model against the rig and quantify the sensitivity of module area to feed temperature.
6. Specify the instrumentation and control philosophy, including the permeate-conductivity interlock for membrane wetting detection.

3.3 Scope and exclusions

IN SCOPE	EXPLICITLY EXCLUDED
Thermal and mass-transfer design of the recovery island	Detailed structural or seismic design of the skid
Heat exchanger sizing and integration with the cooling loop	Data centre IT architecture, server selection, network design
Bench-scale determination of B and rejection	Pilot-scale or field trial (proposed as follow-on work)
Instrumentation schedule and control philosophy	Full control-system implementation and commissioning
Concentrate routing and ZLD interface definition	Design of the crystalliser itself
System-level simulation and model calibration	Certification of permeate for potable supply

4 State of the art

4.1 Data centre cooling architectures

Table 4.1 — Comparison of cooling architectures. Heat capture fraction is the proportion of IT heat delivered to the liquid loop; the balance is handled by the air path.

ARCHITECTURE	LIQUID CAPTURE	RETURN TEMP.	WATER CONSUMPTION	SUITABILITY FOR HEAT RECOVERY
Air cooling + CRAH, chilled water	0 %	12–18 °C	High (evaporative tower)	None — heat is diffuse and below any useful grade
Air cooling + evaporative assist	0 %	25–35 °C	Very high	Marginal
Direct-to-chip, ASHRAE W3	70–80 %	40–45 °C	Moderate	Feasible at low flux
Direct-to-chip, ASHRAE W4 SELECTED	85 %	50–55 °C	Low (dry cooler feasible)	Good — this proposal's baseline
Single-phase immersion	95–100 %	45–60 °C	Very low	Best, but retrofit is not practical

W4 direct-to-chip is selected as the baseline because it achieves a return temperature high enough for thermally-activated separation while remaining a *retrofitable* architecture — cold plates can replace heatsinks in an existing rack, whereas immersion requires a new hall.

4.2 Membrane distillation configurations

CONFIGURATION	PRINCIPLE	TYPICAL GOR	ASSESSMENT FOR THIS DUTY
DCMD — direct contact	Cold permeate in contact with membrane	< 1	Simplest; high conduction loss. Used for the <i>bench rig</i> because it isolates <i>B</i> cleanly.
AGMD — air gap	Stagnant air gap before condensing surface	2–4	Lower conduction loss; internal latent heat recovery.
V-MEMD — multi-effect SELECTED	Latent heat of each effect drives the next	3–6	Selected for the <i>plant</i> design. GOR 4 assumed, giving 165 kWh/m³.
SGMD / VMD	Sweep gas / vacuum on permeate side	varies	Additional blower or vacuum plant; rejected on parasitic power.

ASSUMPTION TO BE STATED OPENLY

The specific thermal energy of 165 kWh/m³ assumes internal latent-heat recovery at GOR 4. Single-pass DCMD without recovery requires 600–900 kWh/m³ and would demand roughly four times the heat duty. The distinction is material and is carried explicitly through §7.5.

4.3 Membrane materials

Hydrophobic microporous membranes of PTFE, PVDF or PP are used. The relevant properties are pore size (0.2–0.45 µm), porosity (70–85 %), thickness (50–200 µm), and liquid entry pressure (LEP), which sets the hydraulic margin before liquid intrusion — *wetting* — destroys selectivity. PTFE is selected for its higher LEP and chemical resistance to the biocides present in cooling-tower blowdown.

4.4 Prior art

A 2018 patent describes waste-heat-driven low-temperature desalination integrated with data centre cooling. It claims a fixed hardware topology on a single steady liquid loop with a seawater feed. It does not address a mixed air/liquid-cooled

fleet, an inland blowdown feed, transient heat availability tracking IT load, or a hybrid pressure-driven/thermally-driven train. The present work is distinguished on all four counts, and in particular on the transient thermal integration, which is where the engineering difficulty actually lies.

5 System description

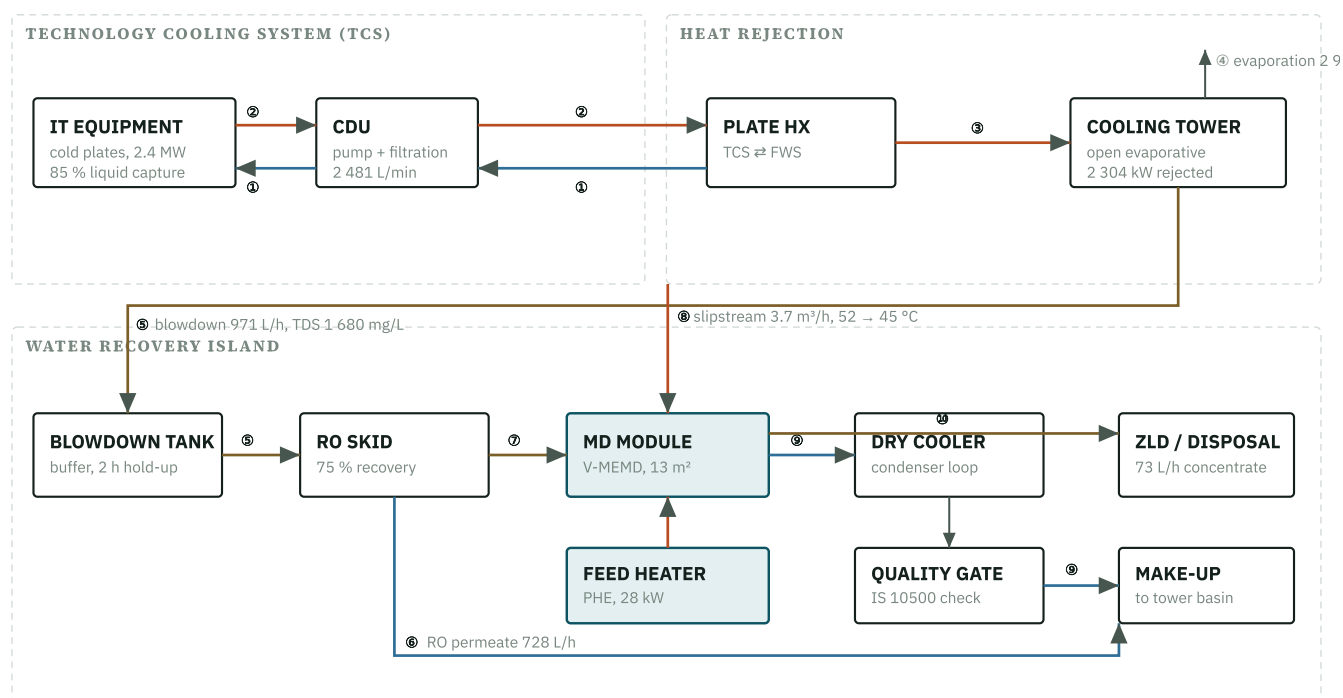


Figure 5.1 — Process flow diagram. Red: hot coolant / recovered heat. Blue: cooled coolant and recovered water. Brown: concentrated aqueous streams. Circled numbers are stream tags referenced in Table 5.1. The MD module and its feed heater (shaded) are the elements this proposal designs; everything else is either existing plant or standard equipment.

Table 5.1 — Stream table at design conditions (2.4 MW IT load, open-tower retrofit case).

NO.	STREAM	FLOW	TEMP. °C	TDS MG/L	F ⁻ MG/L	NOTES
1	TCS coolant supply	2 481 L/min	40.0	—	—	PG25, closed loop
2	TCS coolant return	2 481 L/min	52.0	—	—	ΔT = 12 K
3	FWS to cooling tower	—	48 / 38	1 680	7.2	2 304 kW rejected
4	Evaporation to atmosphere	2 913 L/h	—	0	0	Consumptive loss
5	Tower blowdown	971 L/h	35	1 680	7.2	CoC = 4
6	RO permeate	728 L/h	32	< 60	< 0.3	Returned to basin
7	RO reject → MD feed	243 L/h	32	6 720	28.8	π = 5.2 bar
8	Heat slipstream	3.7 m ³ /h	52 → 45	—	—	2.5 % of TCS flow, 28 kW
9	MD permeate	170 L/h	31	< 40	< 0.2	Gated against IS 10500
10	MD concentrate	73 L/h	42	22 400	96	To ZLD, π = 17.5 bar

6 Thermal analysis

6.1 Energy balance

Taking the IT hall as the control volume, and neglecting the small fraction of electrical energy leaving as radiation and acoustic work:

$$\dot{Q}_{\text{total}} = \eta_{\text{heat}} \cdot P_{\text{IT}} = 0.96 \times 2\,400 = 2\,304 \text{ kW} \quad (6.1)$$

$$\dot{Q}_{\text{liq}} = f_{\text{cap}} \cdot \dot{Q}_{\text{total}} = 0.85 \times 2\,304 = 1\,958 \text{ kW} \quad ; \quad \dot{Q}_{\text{air}} = 346 \text{ kW} \quad (6.2)$$

where f_{cap} is the liquid capture fraction of the cold-plate architecture (Table 4.1). The air-borne balance serves memory, drives and power supplies and is handled by the existing CRAH path.

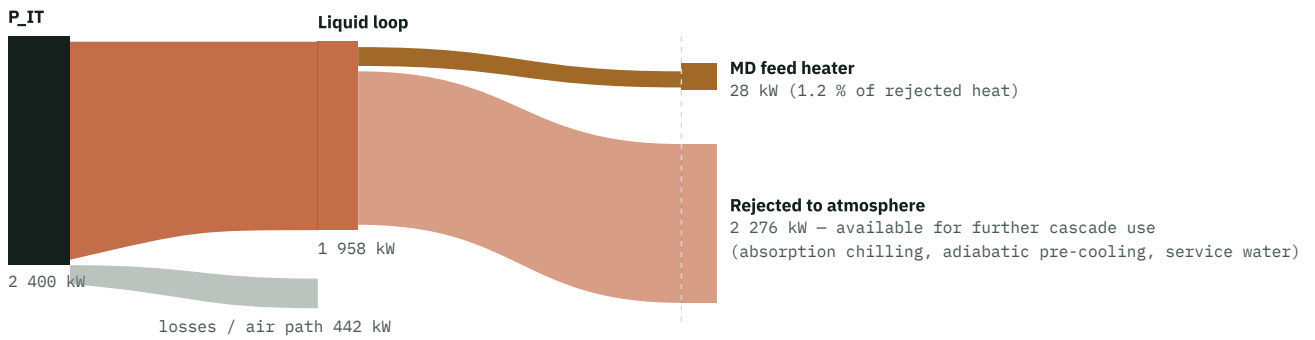


Figure 6.1 — Energy flow. The MD duty is a very small fraction of the rejected heat. The binding constraint is not the *quantity* of waste heat but its *grade* — which is why §6.5 treats the cascade explicitly and why the architecture selection in §4.1 dominates the design.

6.2 Cold plate thermal resistance and junction temperature

The die-to-fluid path is modelled as resistances in series. The junction temperature is the sum of the local coolant temperature and the product of device dissipation and total resistance:

$$T_j = T_{c,\text{local}} + \dot{Q}_{\text{die}} (R_{jc} + R_{\text{TIM}} + R_{sp} + R_{\text{conv}}) \quad (6.3)$$

$$R_{\text{conv}} = 1 / (h \cdot A_{\text{eff}}) \quad ; \quad \text{Nu} = f(\text{Re}, \text{Pr}) \text{ for the microchannel} \quad (6.4)$$

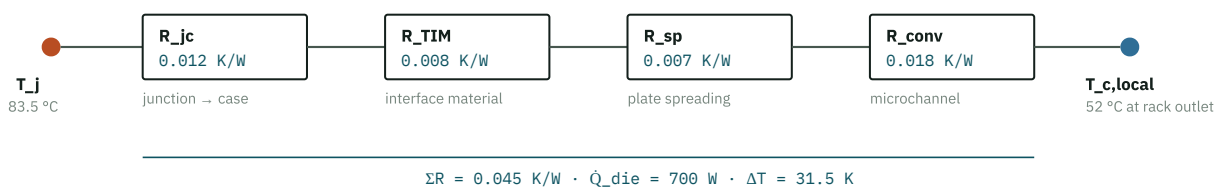


Figure 6.2 — Die-to-coolant resistance network. With the rack-outlet coolant at 52 °C, $T_{j,\text{max}} = 83.5$ °C against a device limit of 90 °C — a margin of 6.5 K. This margin is the price paid for the elevated return temperature and is the principal constraint on the W4 architecture.

DESIGN TENSION TO BE RESOLVED

Raising the coolant supply temperature raises the MD flux (§7.5) but erodes junction-temperature margin. The margin of 6.5 K is workable but not generous; the sensitivity study in §7.5 quantifies exactly what is bought thermally for each kelvin of margin surrendered. Falling back to W3 (supply 32 °C, return 44 °C) restores 8 K of margin at the cost of roughly 45 % of the MD flux.

6.3 Cooling tower water balance

Heat rejection in an open tower is predominantly latent. Taking the latent fraction f_{lat} and the enthalpy of vaporisation at the operating wet-bulb:

$$\dot{m}_{\text{evap}} = f_{\text{lat}} \cdot \dot{Q}_{\text{total}} / h_{\text{fg}} = 0.85 \times 2\,304 / 2\,420 = 0.809 \text{ kg/s} = 2\,913 \text{ L/h} \quad (6.5)$$

Dissolved solids concentrate in the circulating water. The cycles of concentration relate blowdown to evaporation by a solids balance:

$$\text{CoC} = c_{\text{circ}} / c_{\text{makeup}} = (\dot{m}_{\text{evap}} + \dot{m}_{\text{bd}}) / \dot{m}_{\text{bd}} \implies \dot{m}_{\text{bd}} = \dot{m}_{\text{evap}} / (\text{CoC} - 1) \quad (6.6)$$

$$\dot{m}_{\text{bd}} = 2\,913 / 3 = 971 \text{ L/h} ; \quad \dot{m}_{\text{makeup}} = 3\,884 \text{ L/h} = 93.2 \text{ m}^3/\text{d} \quad (6.7)$$

Drift loss is taken at 0.005 % of circulating flow and is neglected in the balance. CoC = 4 is set by the Langelier saturation index of the make-up water; raising it reduces blowdown but increases scaling risk.

6.4 Heat recovery exchanger

A plate heat exchanger transfers the MD duty from a slipstream of the TCS return to the MD feed circuit. Sizing by the log-mean temperature difference:

$$\Delta T_{\text{lm}} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2) = (4 - 3) / \ln(4/3) = 3.48 \text{ K} \quad (6.8)$$

$$UA = \dot{Q} / \Delta T_{\text{lm}} = 28.1 / 3.48 = 8.1 \text{ kW/K} \implies A = UA / U = 2.0 \text{ m}^2 \text{ at } U = 4 \text{ kW/m}^2\text{K} \quad (6.9)$$

Verified by the effectiveness–NTU method for a counter-flow arrangement:

$$\varepsilon = (1 - e^{-NTU(1-C_r)}) / (1 - C_r e^{-NTU(1-C_r)}) = 0.70 \quad \text{at } NTU = 2.01, C_r = 0.86 \quad (6.10)$$

which agrees with $\varepsilon = \dot{Q} / [C_{\text{min}} (T_{\text{h,in}} - T_{\text{c,in}})] = 28.1 / [17.7 \times 10] = 0.70$, confirming the sizing.

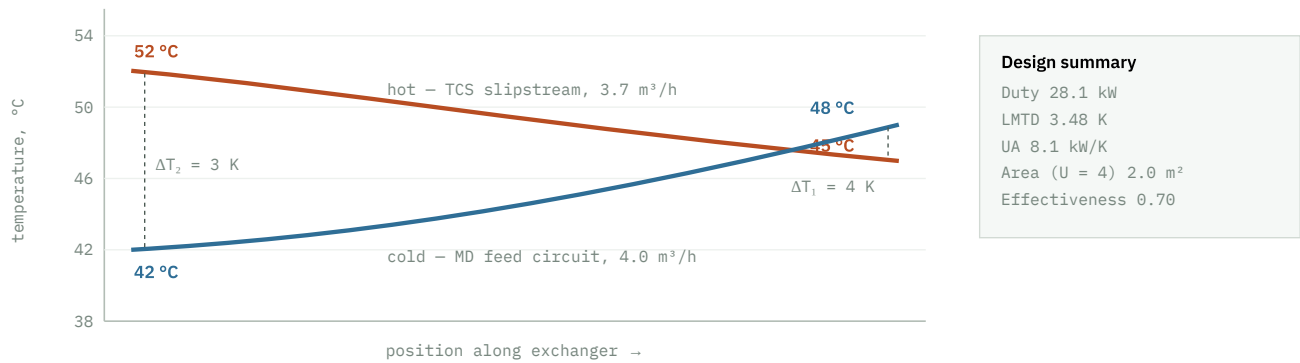


Figure 6.3 — Counter-flow temperature profile in the MD feed heater. The small approach temperatures are characteristic of low-grade recovery: a 3.5 K LMTD demands a large surface area per kilowatt. Counter-flow is essential — a parallel-flow arrangement at these temperatures produces a temperature cross and is infeasible.

6.5 Heat grade cascade

The rejected heat is a single stream but not a single resource: its usefulness is set by temperature, not quantity. Matching each application to the lowest grade that will serve it — a cascade — maximises the total work extracted before the remainder is rejected.

Table 6.1 — Heat grade cascade for the design facility.

GRADE	APPLICATION	COMMENT
65–95 °C	Membrane distillation at high flux; absorption chilling (double effect)	Not available from W4; would require a heat pump lift or W5 operation
45–65 °C	Membrane distillation feed; single-effect absorption chilling; service hot water	The W4 return sits here. Design point of this proposal.
38–45 °C	Marginal MD; adiabatic pre-cooling of the dry cooler; space heating	Driving force collapses below ≈ 38 °C
< 38 °C	Rejection to tower or dry cooler	No further extraction practicable

The MD duty is only 1.2 % of the rejected heat (Fig. 6.1), so cascading is not required to satisfy it. The cascade is included because it establishes the correct framing for the follow-on question — *what else should the remaining 2 276 kW do?* — and because it explains why the architecture decision (which sets grade) matters more than the recovery equipment (which sets quantity).

7 Membrane distillation module design

7.1 Transport mechanism

A hydrophobic microporous membrane separates the hot feed from the cold permeate. Surface tension prevents the aqueous phase from entering the pores provided the transmembrane hydraulic pressure remains below the liquid entry pressure. Water evaporates at the feed-side pore mouth, the vapour is transported across the pore, and condenses on the permeate side.

Because transport occurs entirely in the vapour phase, non-volatile species — Na^+ , Ca^{2+} , Cl^- , SO_4^{2-} , F^- and NO_3^- — are excluded by the phase change itself rather than by size exclusion. This is the property the proposal seeks to quantify experimentally.

Vapour transport through the pore is described by a combined Knudsen–molecular diffusion model. The controlling regime is set by the Knudsen number, $\text{Kn} = \lambda / d_p$; for $d_p = 0.2 \mu\text{m}$ and a mean free path of $\approx 0.11 \mu\text{m}$ at operating conditions, $\text{Kn} \approx 0.55$, placing the module in the transition regime where both mechanisms contribute. For design purposes the transport is lumped into a single membrane distillation coefficient:

$$J = B (p_w(T_{fm}) \cdot a_w - p_w(T_{pm})) \quad (7.1)$$

where J is permeate flux ($\text{L m}^{-2} \text{h}^{-1}$), B the membrane distillation coefficient ($\text{L m}^{-2} \text{h}^{-1} \text{Pa}^{-1}$), p_w the saturation vapour pressure and a_w the water activity of the feed. Determining B for the selected membrane at this grade of heat is **Objective 4** of the proposal.

7.2 Driving force

Saturation vapour pressure is evaluated from the Antoine correlation:

$$\log_{10} p_w = 8.07131 - 1730.63 / (233.426 + T) \quad [p_w \text{ in mmHg, } T \text{ in } ^\circ\text{C}] \quad (7.2)$$

Water activity is reduced by the dissolved solids. Even at the MD concentrate condition of $22\,400 \text{ mg/L}$, $a_w \approx 0.992$ — a flux penalty of 0.8 %, against a thirteen-fold rise in osmotic pressure over the same range (Table 2.1). The exponential temperature dependence of p_w therefore dominates the design entirely.

7.3 Temperature polarisation

Thermal boundary layers form on both sides of the membrane. The temperatures that matter are the *membrane surface* temperatures, not the bulk values an operator measures. Defining the temperature polarisation coefficient:

$$\tau = (T_{fm} - T_{pm}) / (T_f - T_p) \quad ; \quad T_{fm} = T_f - \frac{1}{2}(1-\tau)(T_f - T_p) \quad (7.3)$$

A value of $\tau = 0.75$ is assumed for design, consistent with spacer-filled channels at moderate Reynolds number. Neglecting polarisation entirely ($\tau = 1$) would overstate the flux by approximately 30 % — an error that propagates directly into an undersized module.

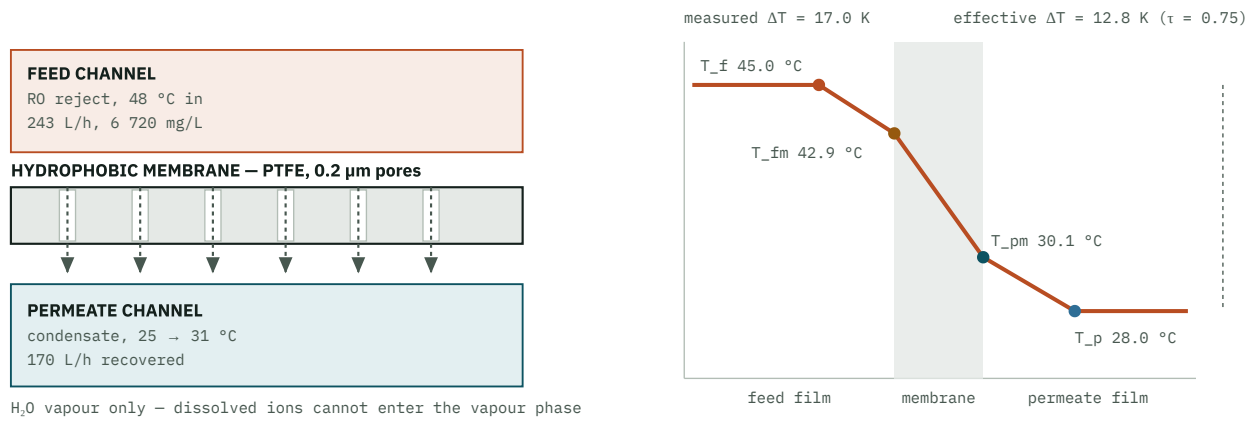


Figure 7.1 — Module cross-section and temperature profile. Left: the direct-contact arrangement, with vapour crossing the pores while dissolved species remain in the feed. Right: the temperature profile across both boundary layers. The driving force available to the membrane is 12.8 K, not the 17.0 K measured between the bulk streams.

7.4 Concentration polarisation, scaling and wetting

- **Concentration polarisation** raises the feed-side wall concentration above the bulk. Its effect on flux through a_w is small (§7.2) but it governs the local saturation index and therefore the scaling risk, which is the practical limit on recovery.
- **Scaling.** CaCO₃ and CaSO₄ are the controlling species; silica becomes limiting above roughly 150 mg/L SiO₂. Mitigation: antiscalant dosing to the RO feed, Langelier index monitoring, and periodic acid clean-in-place.
- **Wetting** is the dominant failure mode. Liquid intrusion into the pores destroys the vapour barrier and ions convect through directly. It is detected as a step rise in permeate conductivity — this is the basis for the interlock specified in §9.2 and the failure condition modelled in the simulation (§10).

7.5 Module sizing and sensitivity to feed temperature

Required area follows from the target permeate rate and the flux at the design point:

$$A_m = \dot{V}_{perm} / J = 170 / 12.9 = 13.2 \text{ m}^2 \Rightarrow \text{specify } 15 \text{ m}^2 \text{ installed} \quad (7.4)$$

Table 7.1 — Flux and required membrane area against MD feed inlet temperature. $B = 3.0 \times 10^{-3} \text{ L m}^{-2} \text{ h}^{-1} \text{ Pa}^{-1}$, $\tau = 0.75$, permeate bulk mean 28 °C, permeate target 170 L/h.

$T_{F,IN}$ °C	T_{FM} °C	T_{PM} °C	ΔP KPA	J L M ⁻² H ⁻¹	A_m M ²	SOURCE OF HEAT
40	35.9	29.1	1.86	5.6	30.4	W3 return, marginal
45	40.3	29.8	3.29	9.9	17.2	W3 return
48	42.9	30.1	4.30	12.9	13.2	Design point — W4 return
52	46.4	30.6	5.87	17.6	9.6	W4, smaller HX approach
55	49.0	31.0	7.23	21.7	7.8	W5 return
60	53.4	31.6	9.88	29.6	5.7	Requires heat pump lift

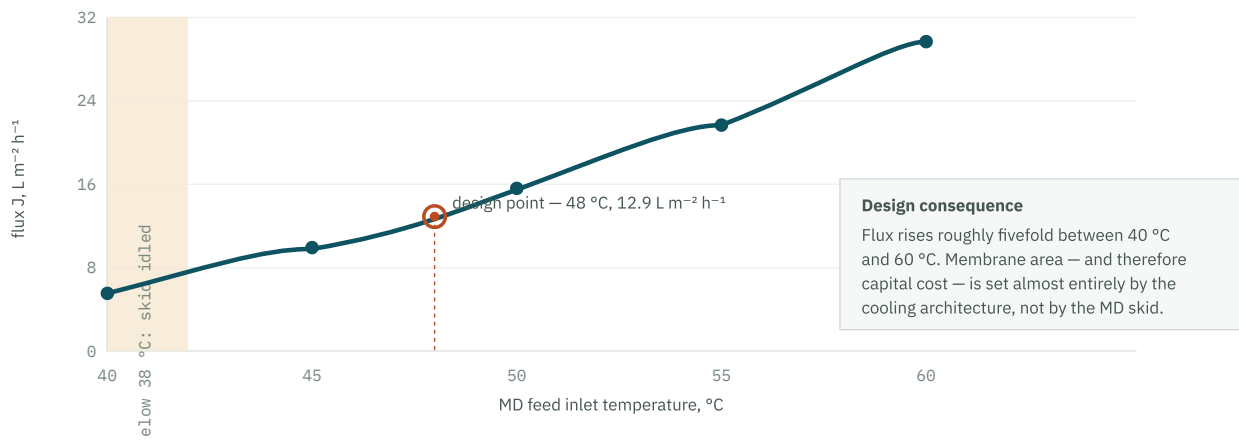


Figure 7.2 — Flux against feed temperature. The strong non-linearity follows from the Clausius–Clapeyron behaviour embedded in Eq. (7.2). It is the quantitative basis for selecting a W4 rather than W3 cooling architecture, and for the design tension identified in §6.2.

7.6 Concentrate management

The MD concentrate — 73 L/h at approximately 22 400 mg/L — is a smaller but more concentrated stream than the blowdown it originated from. It is routed to a crystalliser for zero liquid discharge, or to licensed industrial reuse. Its design is outside the scope of this project (§3.3), but the interface conditions are specified so that the recovery island can be tendered against a defined concentrate duty. A recovery scheme that does not account for its own residual stream is not a closed design.

8 Integrated design case

8.1 Summary of the design point

Table 8.1 — Consolidated design summary, 2.4 MW IT facility, open-tower retrofit.

PARAMETER	VALUE	BASIS
IT load	2 400 kW	Design assumption
Total heat rejected	2 304 kW	Eq. (6.1)
Liquid-captured heat	1 958 kW	Eq. (6.2), $f_{\text{cap}} = 0.85$
Coolant supply / return	40 / 52 °C	ASHRAE W4
Coolant flow rate	2 481 L/min	$\dot{Q} = \dot{m} c_p \Delta T$, PG25
Maximum junction temperature	83.5 °C	Eq. (6.3); 6.5 K margin
Tower evaporation	2 913 L/h	Eq. (6.5)
Tower blowdown, CoC = 4	971 L/h	Eq. (6.6)
RO permeate / reject	728 / 243 L/h	75 % recovery
MD permeate / concentrate	170 / 73 L/h	70 % recovery
Overall blowdown recovery	92.5 %	898 of 971 L/h
MD thermal duty	28.1 kW	165 kWh/m ³ at GOR 4
Membrane area installed	15 m ²	Eq. (7.4) + margin
Feed heater area	2.0 m ²	Eq. (6.9)
Parasitic electrical load	1.7 kW	Pumps + dry cooler fan
Specific electrical energy	10.3 kWh/m ³	Of MD permeate

8.2 Water balance under three architectures

The recovery island addresses the blowdown; the cooling architecture addresses the evaporation. Because evaporation is three quarters of the make-up demand, the two measures are not alternatives — the architectural change dominates, and recovery closes the remainder.

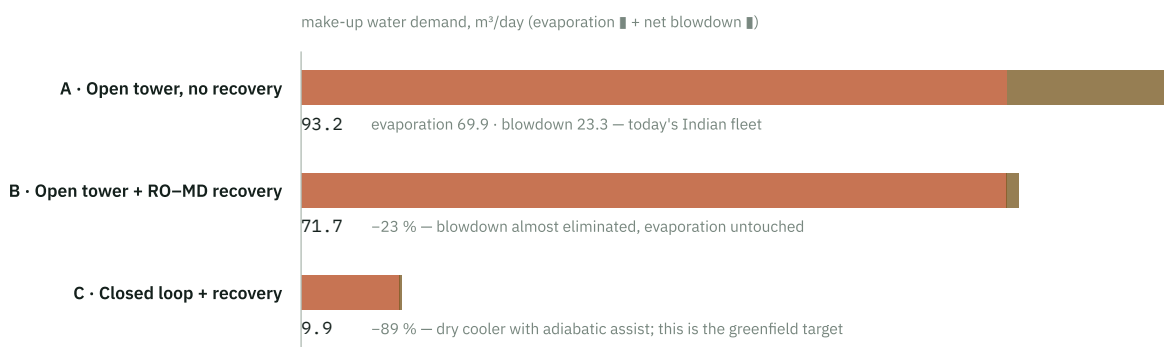


Figure 8.1 — Make-up water demand under three cases. Recovery alone (B) removes 23 % of demand because it can only address the blowdown. Architectural change plus recovery (C) removes 89 %. The engineering conclusion is that *the recovery island is a retrofit measure for the existing evaporative fleet*, while new-build facilities should adopt closed-loop rejection and use the recovered heat elsewhere in the cascade.

8.3 Alternative configuration: MD without RO

An MD-only train treating the full blowdown was evaluated as an alternative. At 75 % recovery it yields 728 L/h of permeate, requires 120 kW of heat and 56 m² of membrane — roughly four times the area and duty of the hybrid for the same total water recovery, because MD is applied to a dilute stream where RO is far cheaper. It is retained only as a fallback for sites where the blowdown chemistry (high silica, high hardness) makes RO pretreatment uneconomic.

9 Instrumentation and control

9.1 Instrumentation schedule

Table 9.1 — Instrumentation for the recovery island and bench rig.

TAG	MEASUREMENT	TYPE	RANGE	ACCURACY	PURPOSE
TT-01/02	Coolant supply / return	RTD Pt100 Class A	0–100 °C	± 0.15 K	Heat duty, cascade band
TT-03/04	MD feed in / out	RTD Pt100 Class A	0–100 °C	± 0.15 K	Driving force, Eq. (7.1)
TT-05/06	Permeate in / out	RTD Pt100 Class A	0–100 °C	± 0.15 K	Condenser duty
FT-01	Coolant flow	Electromagnetic	0–3 000 L/min	± 0.5 % rdg	Energy balance closure
FT-02/03	Blowdown, permeate	Electromagnetic	0–2 000 L/h	± 0.5 % rdg	Flux, mass balance
CT-01/02	Feed / permeate conductivity	Toroidal, 4-electrode	0–50 mS/cm	± 1 % rdg	Wetting detection interlock
AT-01	Fluoride	Ion-selective electrode	0.02–100 mg/L	± 5 %	IS 10500 compliance
AT-02	Nitrate	Ion-selective electrode	0.1–1 000 mg/L	± 5 %	IS 10500 compliance
AT-03	pH	Glass electrode	0–14	± 0.1	Scaling index, compliance
PDT-01	Transmembrane Δp	Differential pressure	0–1 bar	± 0.25 % FS	LEP margin, fouling trend
JT-01/03	Pump / fan power	3-phase power meter	0–5 kW	± 1 %	Parasitic load, SEC

KEY MEASUREMENT

Permeate conductivity (CT-02) is the primary integrity measurement. Intact membrane operation yields a permeate conductivity two to three orders of magnitude below the feed. A step increase is the unambiguous signature of pore wetting, and it precedes any detectable rise in fluoride at the ISE. It is therefore used as the interlock rather than the fluoride probe, which is slower and requires more frequent calibration.

9.2 Control philosophy

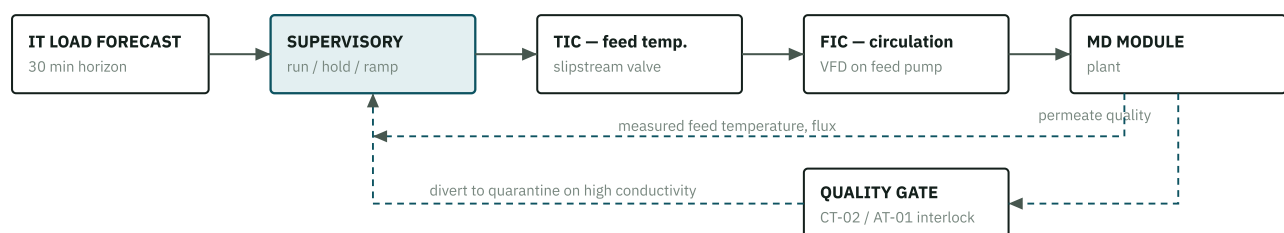


Figure 9.1 — Control block diagram. Feedback control on feed temperature and circulation rate; feed-forward from the IT load forecast so the skid ramps ahead of a change in heat availability rather than chasing it. The quality gate is an interlock, not a controller: it removes product from the delivery path rather than adjusting a setpoint.

- **Regulatory layer.** TIC-01 modulates the slipstream control valve to hold MD feed inlet at setpoint; FIC-01 sets feed circulation to maintain channel Reynolds number and hence τ .
- **Supervisory layer.** Run/hold decision on available return temperature and buffer tank level. Below 38 °C feed the driving force does not justify parasitic power and the skid is idled.
- **Feed-forward.** A short-horizon IT load forecast anticipates heat availability, allowing the skid to pre-ramp. This is where the software layer described in §10 sits.
- **Interlocks.** High permeate conductivity → divert to quarantine; low LEP margin on PDT-01 → reduce feed pressure; high LSI → increase blowdown or dose antiscalant.

10 System simulation software

A real-time system model has been implemented as a working console. Its purpose is engineering: it propagates a change at any point in the chain through every downstream calculation, so the coupling between architecture, heat grade, flux and permeate chemistry can be examined interactively rather than through static spreadsheets.

The model implements the equations of §6 and §7 directly — Eq. (6.1) through (6.7) for the thermal and water balance, Eq. (7.1) to (7.3) for the membrane, and the concentration-then-rejection chain for permeate chemistry. It is not an animation; every displayed quantity is computed.

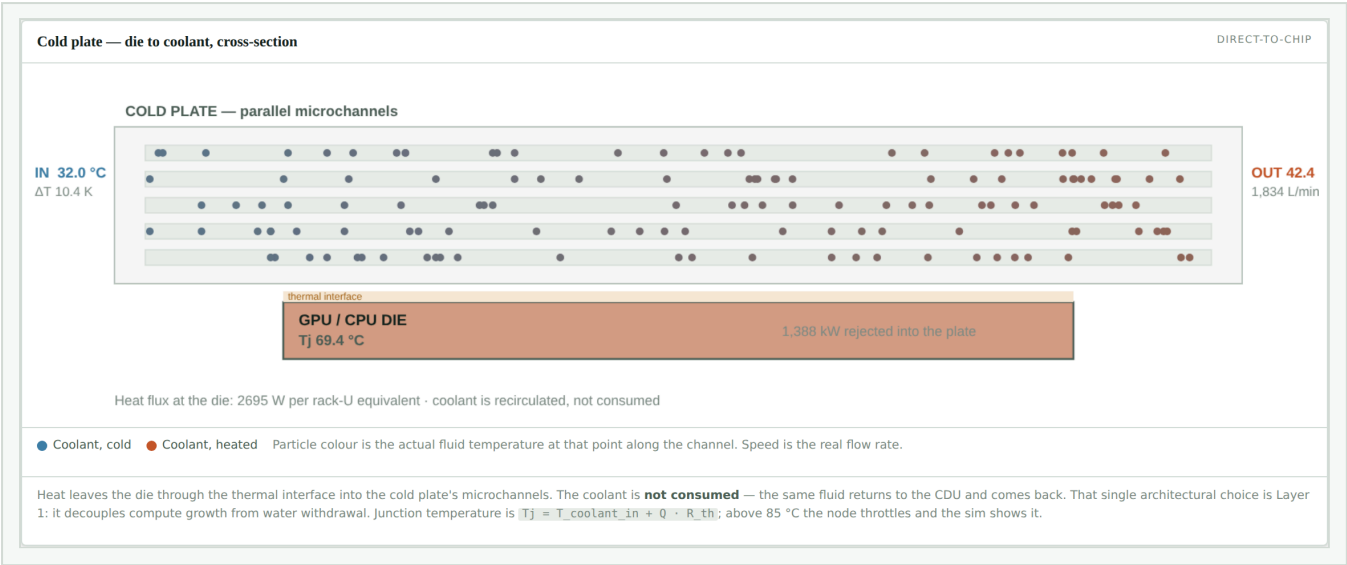


Figure 10.1 — Cold plate module. Coolant elements are coloured by the local fluid temperature interpolated between supply and return, and advected at the true volumetric flow rate. Junction temperature is evaluated from Eq. (6.3); the throttle condition is flagged when T_j exceeds 85 °C.

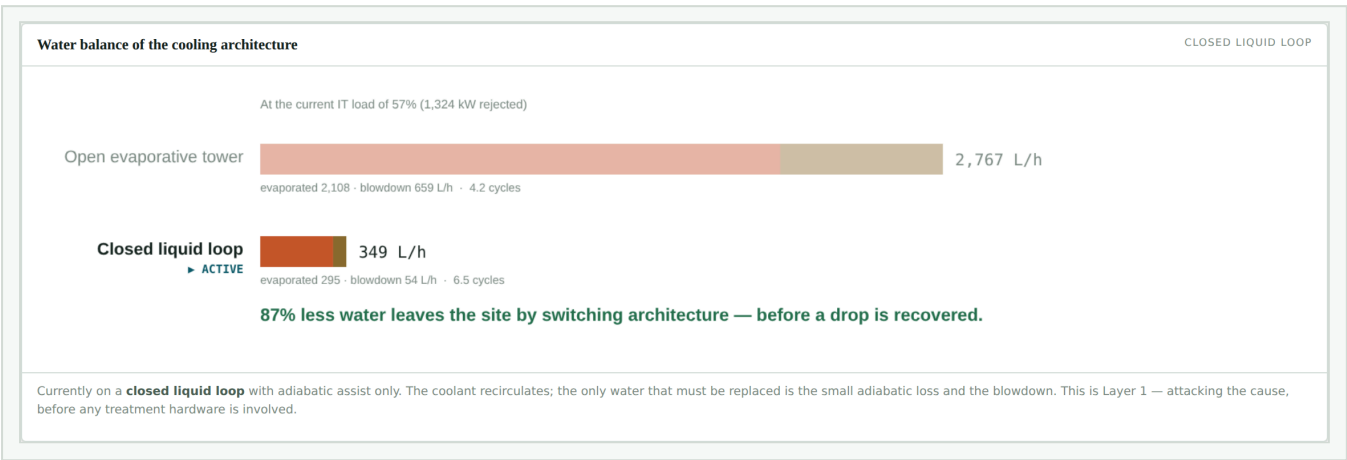


Figure 10.2 — Architecture water balance. Evaporation and blowdown per hour at the current heat load, computed from Eq. (6.5) and (6.6), for open tower against closed loop. Switching architecture in the model reproduces the Case A → Case C comparison of Fig. 8.1.

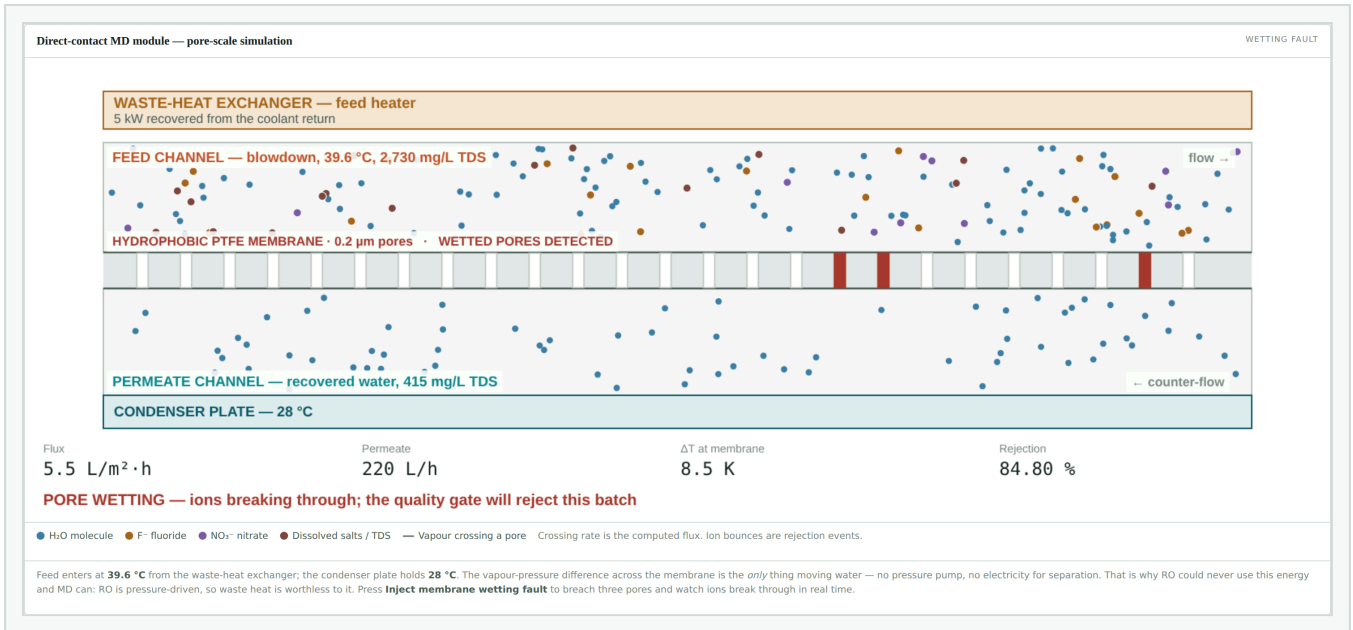


Figure 10.3 — Membrane module. Vapour transits across the pores occur at a rate proportional to the computed flux; dissolved species are rejected at the membrane face. Shown here **DURING AN INJECTED WETTING FAULT** : three pores breached, ion breakthrough visible, rejection fallen from 99.4 % to 84.8 %. This is the failure mode the CT-02 interlock (§9.1) is specified to detect.

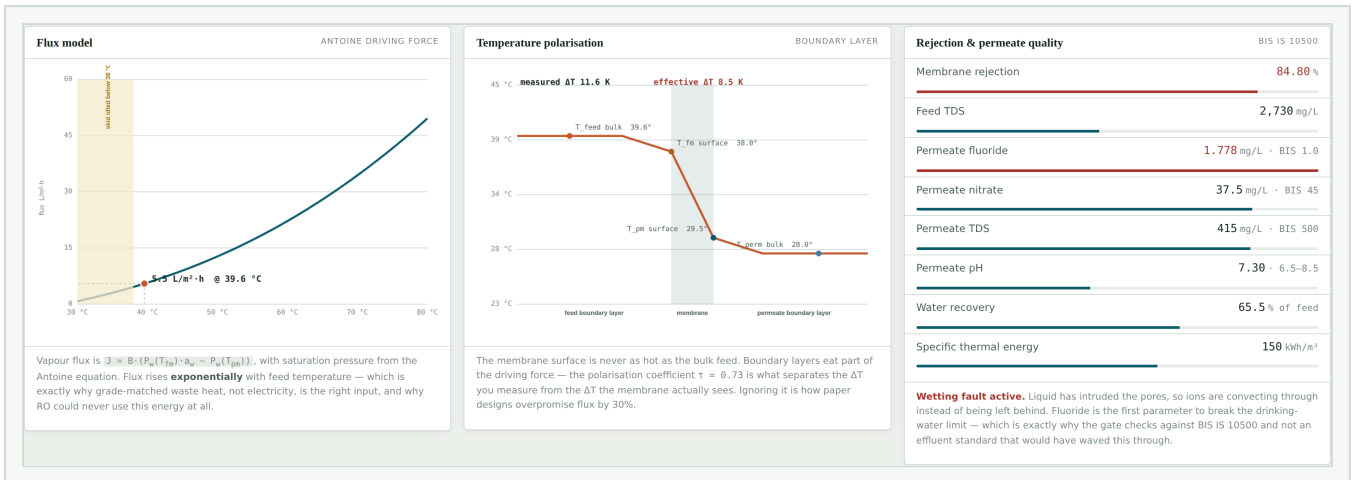


Figure 10.4 — Transport model outputs. Left: the flux curve of Fig. 7.2 with the live operating point. Centre: the temperature profile of Fig. 7.1, distinguishing measured from effective ΔT. Right: permeate chemistry against IS 10500, with fluoride in breach under the injected fault.

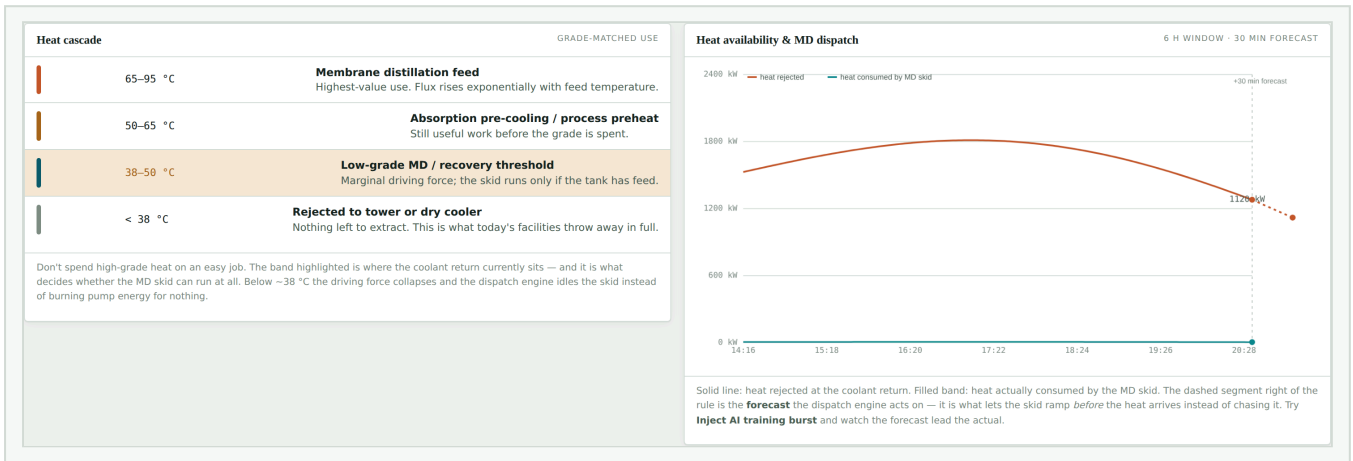


Figure 10.5 — Heat cascade and dispatch. Left: the grade bands of Table 6.1, with the current coolant return highlighted. Right: heat rejected against heat consumed by the recovery island, with a 30-minute forecast — the feed-forward signal of Fig. 9.1.

11 Experimental validation plan

11.1 Objective of the experimental work

Two quantities in the design are taken from literature and must be measured for the specific membrane and feed: the membrane distillation coefficient B in Eq. (7.1), and the ion rejection — particularly for fluoride — at the operating temperatures of interest. A third, the polarisation coefficient τ , is inferred from the difference between measured flux and the flux predicted from bulk temperatures.

11.2 Bench rig

A flat-sheet direct-contact MD cell of 0.5–1.0 m² effective area, with counter-current feed and permeate channels separated by the test membrane. DCMD is selected for the rig rather than the multi-effect configuration of the plant because it isolates the single membrane transport step cleanly, which is what B describes.

- Two thermostatic circulators, ± 0.1 K stability, setting feed and permeate bulk temperatures.
- Gear pumps with VFD for independent channel velocity control.
- Permeate collected on a balance; flux by gravimetric rate, cross-checked against FT-03.
- Instrumentation per Table 9.1, logged at 1 Hz.
- Synthetic feed made up to Devanahalli blowdown chemistry: TDS 1 680 mg/L, F[−] 7.2 mg/L, NO₃[−] 152 mg/L, hardness as CaCO₃ 480 mg/L; and a concentrated variant at 6 720 mg/L representing the RO reject.

11.3 Test matrix

SERIES	VARIABLE	LEVELS	HELD CONSTANT	MEASURED RESPONSE
A — driving force	Feed bulk temperature	40, 45, 50, 55, 60 °C	$T_p = 25$ °C, $v = 0.15$ m/s	Flux → regression for B
B — polarisation	Channel velocity	0.05, 0.10, 0.15, 0.25 m/s	$T_f = 50$ °C	Flux → inferred τ
C — concentration	Concentration factor	1×, 2×, 4×, 8×, 10×	$T_f = 50$ °C	Flux decline, scaling onset
D — rejection	Feed chemistry	2 variants	$T_f = 50$ °C	Permeate F [−] , NO ₃ [−] , conductivity
E — endurance	Time	100 h continuous	Design point	Flux decline, wetting onset
F — membrane	Material / pore size	PTFE 0.2, PTFE 0.45, PVDF 0.22	$T_f = 50$ °C	B , LEP, contact angle

11.4 Uncertainty analysis

Flux is determined gravimetrically, $J = m / (\rho A_m t)$. Propagating the component uncertainties by the root-sum-square method:

$$(u_J/J)^2 = (u_m/m)^2 + (u_A/A)^2 + (u_t/t)^2 \Rightarrow u_J/J \approx \pm 2.6 \% \quad (11.1)$$

The driving force carries the temperature uncertainty through the Antoine derivative, $\partial p_w / \partial T \approx 0.30$ kPa/K at 45 °C. With $u_T = \pm 0.15$ K on each of four RTDs, the propagated uncertainty in Δp is approximately ± 2.4 %, giving a combined uncertainty in B of about ± 3.6 %. Repeatability across three runs is to be reported separately.

11.5 Acceptance criteria

CRITERION	TARGET	CONSEQUENCE IF NOT MET
Measured B against assumed 3.0×10^{-3}	within $\pm 20 \%$	Re-size module area per Eq. (7.4); design method unaffected
Fluoride rejection at design point	$> 99 \%$	Additional polishing stage required before reuse
Flux decline over 100 h	$< 15 \%$	Revise antiscalant regime and CIP interval
Energy balance closure on the rig	$\pm 5 \%$	Instrumentation or insulation fault; re-commission
Inferred τ against assumed 0.75	within ± 0.08	Revise spacer geometry or channel velocity

12 Deliverables, timeline and budget

12.1 Deliverables

1. Validated thermal design method and sizing procedure for a waste-heat-driven MD island, applicable to any facility given its heat-rejection profile and blowdown chemistry.
2. Experimentally determined B , τ and ion rejection for the selected membrane on Devanahalli-chemistry feed.
3. Calibrated system simulation model with a documented equation set.
4. Process flow diagram, stream table, equipment datasheets and instrumentation schedule at design-basis level.
5. Project report and a conference-standard paper on low-grade MD design.

12.2 Timeline

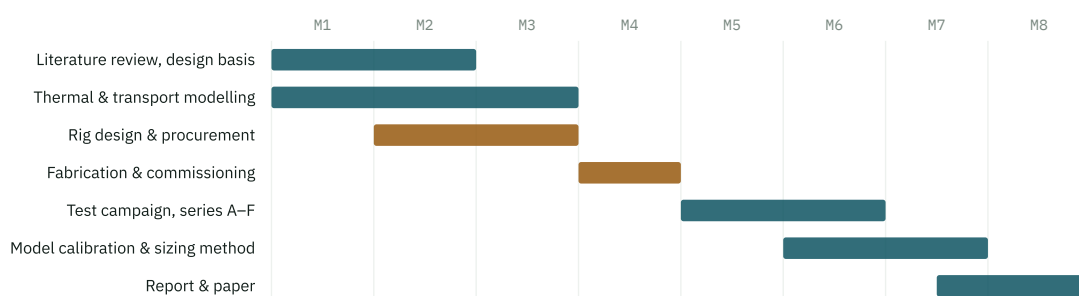


Figure 12.1 — Project schedule. Blue: analytical and modelling work. Amber: hardware. The critical path runs through rig procurement; long-lead items are the membranes and the thermostatic circulators, ordered in M2.

12.3 Bill of materials and budget

Table 12.1 — Indicative budget.

ITEM	SPECIFICATION	COST (₹)
Flat-sheet MD test cell	Stainless, spacer-filled channels, 0.5–1.0 m ²	1,80,000
Membrane coupons	PTFE 0.2 / 0.45 μ m, PVDF 0.22 μ m	45,000
Thermostatic circulators (2)	$\pm$ 0.1 K stability, 2 kW heating	2,20,000
Gear pumps with VFD (2)	0–10 L/min, chemically resistant	60,000
RTDs, flow meters, DAQ	Per Table 9.1; 16-channel logger	2,40,000
Fluoride / nitrate ISE and meter	With calibration standards	95,000
Analytical balance, chemicals	0.01 g, reagent-grade salts, antiscalant	70,000
Fabrication, piping, insulation	Workshop time, fittings, lagging	80,000
Contingency	$\approx$ 6 %	55,000
Total		9,45,000

13 Risk register

RISK	LIKEL.	IMPACT	MITIGATION
Membrane wetting during endurance testing	Medium	High	LEP-verified membranes; feed degassing; permeate conductivity interlock; spare coupons budgeted
Scaling at high concentration factor (Series C)	High	Medium	Antiscalant dosing; LSI monitoring; acid CIP between runs; accept a lower recovery limit as a result
Measured B substantially below literature	Medium	Medium	Module area is a linear function of B (Eq. 7.4); the design method is unaffected, only the sizing
τ substantially below 0.75	Medium	Medium	Series B quantifies it directly; mitigate with spacer redesign or higher channel velocity, at a pumping-power cost
Junction temperature margin unacceptable to an operator at W4	Medium	High	W3 fallback quantified in Table 7.1: 45 % flux penalty, 17.2 m ² area. Design remains feasible
Long-lead procurement delay	Medium	Medium	Order membranes and circulators in M2; modelling work (M1–M4) is off the critical path
Concentrate disposal route unavailable at a host site	Low	High	Interface conditions specified in §7.6 so the duty can be tendered separately; out of project scope

14 Nomenclature and references

14.1 Nomenclature

A_m	Membrane area, m ²	NTU	Number of transfer units, –
a_w	Water activity, –	p_w	Saturation vapour pressure of water, Pa
B	Membrane distillation coefficient, L m ⁻² h ⁻¹ Pa ⁻¹	$\dot{Q}$	Heat transfer rate, kW
C_T	Heat capacity rate ratio, –	R	Thermal resistance, K W ⁻¹
c_p	Specific heat capacity, kJ kg ⁻¹ K ⁻¹	T_{fm}, T_{pm}	Membrane surface temperatures, °C
CoC	Cycles of concentration, –	T_j	Junction temperature, °C
GOR	Gained output ratio, –	U	Overall heat transfer coefficient, kW m ⁻² K ⁻¹
h_{fg}	Enthalpy of vaporisation, kJ kg ⁻¹	ΔT_{1m}	Log mean temperature difference, K
J	Permeate flux, L m ⁻² h ⁻¹	ϵ	Heat exchanger effectiveness, –
Kn	Knudsen number, –	τ	Temperature polarisation coefficient, –

14.2 References

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STATEMENT ON THE PRESENT STATE OF THE WORK

The thermal and mass-transfer model described in §6 and §7 is complete and implemented. All quantities reported in this proposal are computed from that model using property data and coefficients taken from the literature cited above. **No experimental measurements have yet been taken.** The purpose of the work proposed here is to replace the two assumed coefficients — B and τ — and the assumed ion rejection with measured values, and to establish the validity of the sizing method at the low heat grade characteristic of data centre waste heat.